

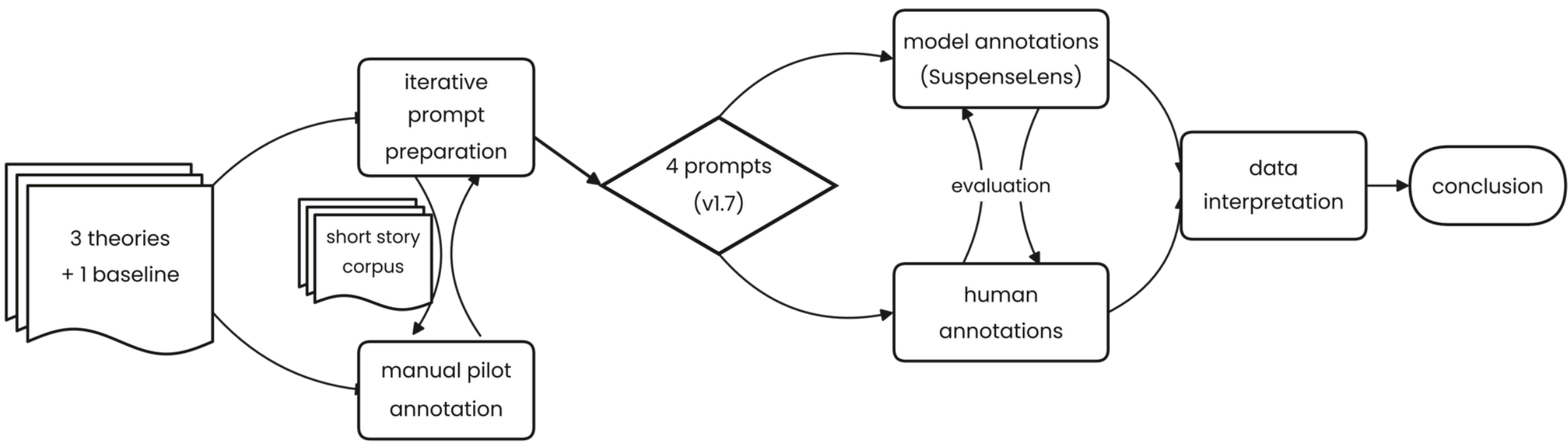
TESTING THEORY PLURALISM AGAINST
LLMS’ INTRINSIC CONCEPTS

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RESEARCH QUESTION AND HYPOTHESIS

Q: Can theory-specific prompts steer the suspense annotations of large language model systems toward a given theory, away from the model's intrinsic, everyday concept of suspense?

H: Theory-specific prompts shift the annotations toward the targeted theory, but the everyday concept (T4), which approximates the intrinsic concept of large language model systems, remains dominant across all conditions and resists being overridden.



3 THEORIES, 1 BASELINE, 4 PROMPTS

System prompt: You are a computational literary studies annotator. For each sentence, annotate **reader suspense** [...] using the selected suspense theory. Rate [it] on a **Likert scale** (0 = no suspense, 5 = very high suspense). [...] Use feed-forward context. Annotate only where reasonable for the selected theory; expect long stretches rated 0. Output **ONLY** a valid **JSON** array — no preamble, no markdown: {'id': <int>, 'reader_suspense_level': <0-5>, [...]} Include sentence IDs. Do not reproduce text.

T1: Uncertainty-based (Nomikos et al. 1968; Iwata 2009)	T2: Desire-frustration (Smuts 2008)	T3: Partial uncertainty (Gerrig 1989)	T4: Casual reader (atheoretical baseline)
Suspense arises from uncertainty about how an event will turn out. Instruction: pay attention to open questions; high suspense where the resolution is open and the feared outcome seems likely.	Suspense arises from a desire to change an outcome the reader knows about but cannot influence. Instruction: read the story twice; high suspense where the reader knows what a character does not.	Suspense persists with outcome knowledge while the manner or timing stays uncertain. Instruction: read the last 10% first; high suspense where an obstacle delays the known outcome.	Everyday suspense from surface cues of tension and threat, independent of any theoretical framework. Instruction: read as a casual reader.

INTER-ANNOTATOR DISAGREEMENT IN HOW IT HAPPENED

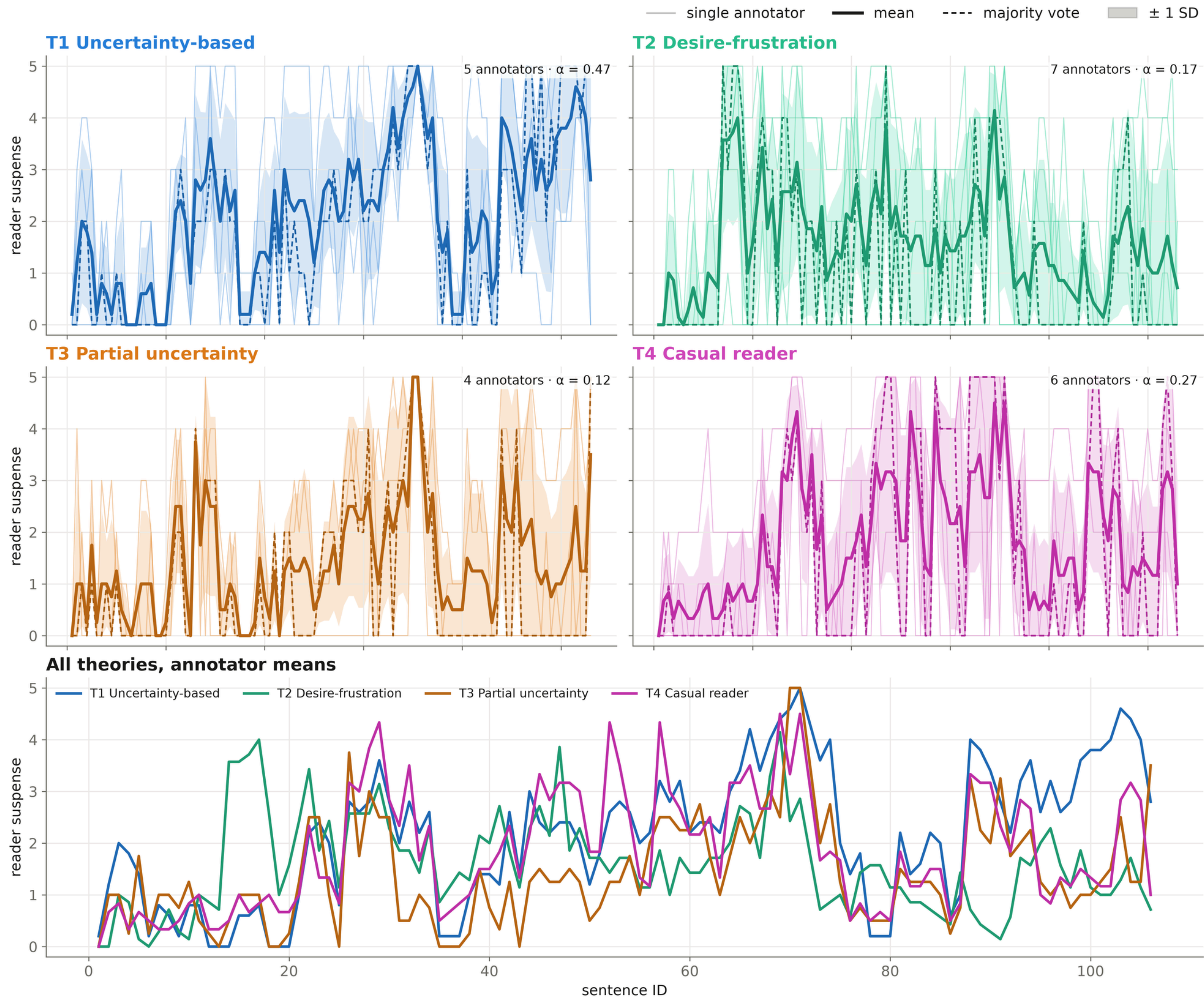


Fig. 1: Reader suspense per sentence and theory: annotators (thin), mean (bold), majority vote (dashed), ±1 SD (band); α = Krippendorff's alpha.

RESULTS

- **Humans disagree by theory.** Agreement ranges from $\alpha = .12$ (T3) to .47 (T1).
- The arcs of T1, T3, and T4 run closely together, while T2 follows its own course (Fig. 1).
- **Models match the casual reader best.** The three strongest models reach $\rho = .61-.67$ on T4, which is consistent with our hypothesis (Fig. 3).
- **T2 stays low.** Agreement with the desire-frustration annotations is weak across models ($\rho \leq .26$).
- **Size does not always help.** Mean ρ rises from .16 (Olmo 3 7B) to .46 (Qwen3.8-27B). In contrast, Gemma 4 31B, the largest model, reaches only .39.

REFERENCES, CODE, AND DATA

Doyle, A. C. 1913. "How It Happened." The Strand Magazine, September 1913.
Gerrig, R. J. 1989. "Suspense in the Absence of Uncertainty." Journal of Memory and Language 28 (6): 633–48. 10.1016/0749-596X(89)90001-6.
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Iwata, Y. 2009. "Creating Suspense and Surprise in Short Literary Fiction: A Stylistic and Narratological Approach." PhD diss., University of Birmingham.
Nomikos, M. S., Opton, E. M., Averill, J. R., and Lazarus, R. S. 1968. "Surprise versus Suspense in the Production of Stress Reaction." Journal of Personality and Social Psychology 8 (2): 204–8.
Smuts, A. 2008. "The Desire-Frustration Theory of Suspense." Journal of Aesthetics and Art Criticism 66 (3): 281–90. 10.1111/j.1540-6245.2008.00309.x.

SUSPENSE LENS

- Browser tool connected to LM Studio for running open-weight models locally
- Uses the same four prompts as the human annotators
- Feed-forward reading: the model annotates each sentence with knowledge of the story so far
- Annotations color-coded and filterable by suspense level and theory
- Compared models:

OLMO 3 7B QWEN3-4B GEMMA 4 31B
LLAMA 3.1 8B QWEN3.8-27B

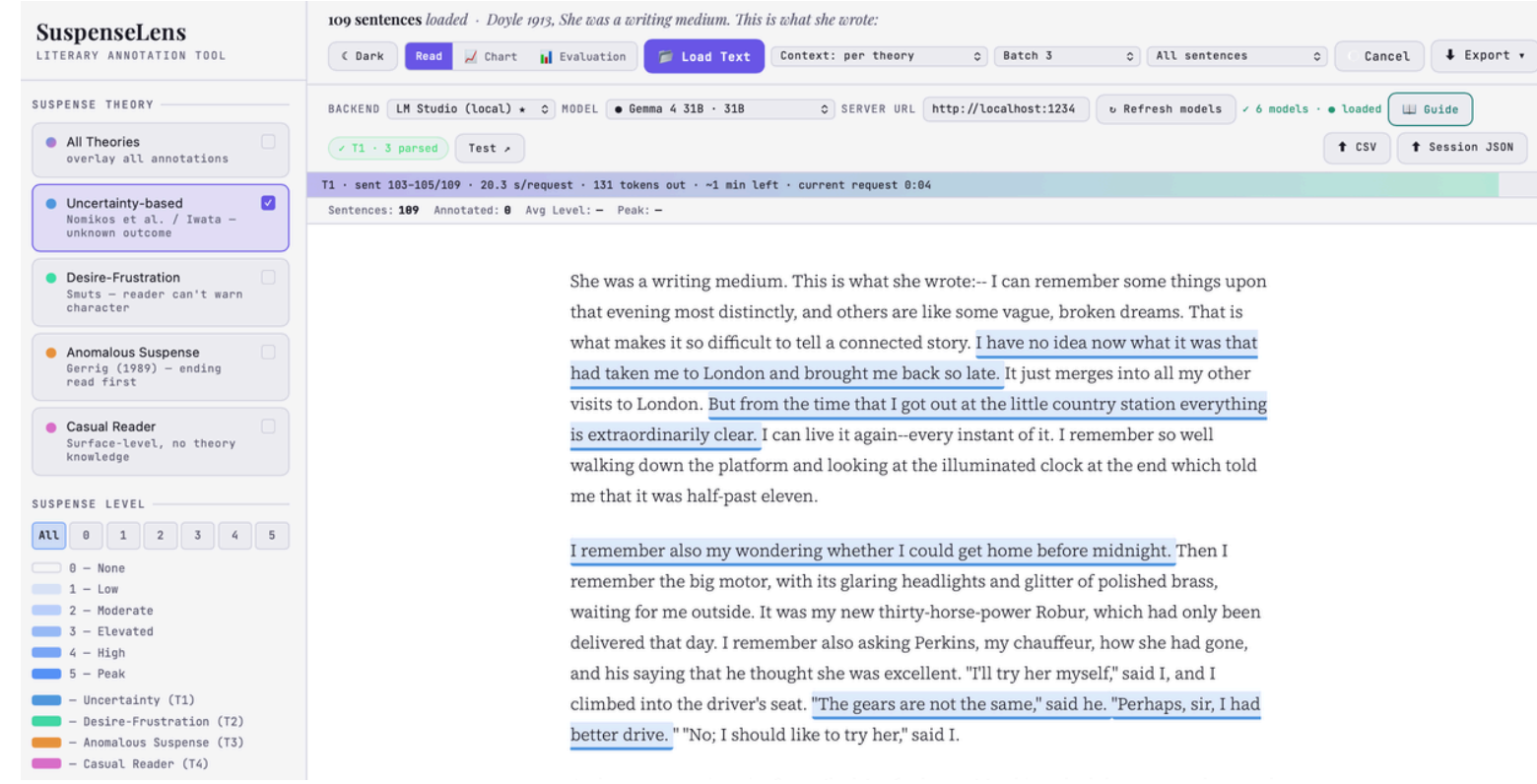


Fig. 2: SuspenseLens during a T1 run of Gemma 4 31B on "How It Happened". Shading marks the reader suspense level of annotated sentences.

EVALUATION

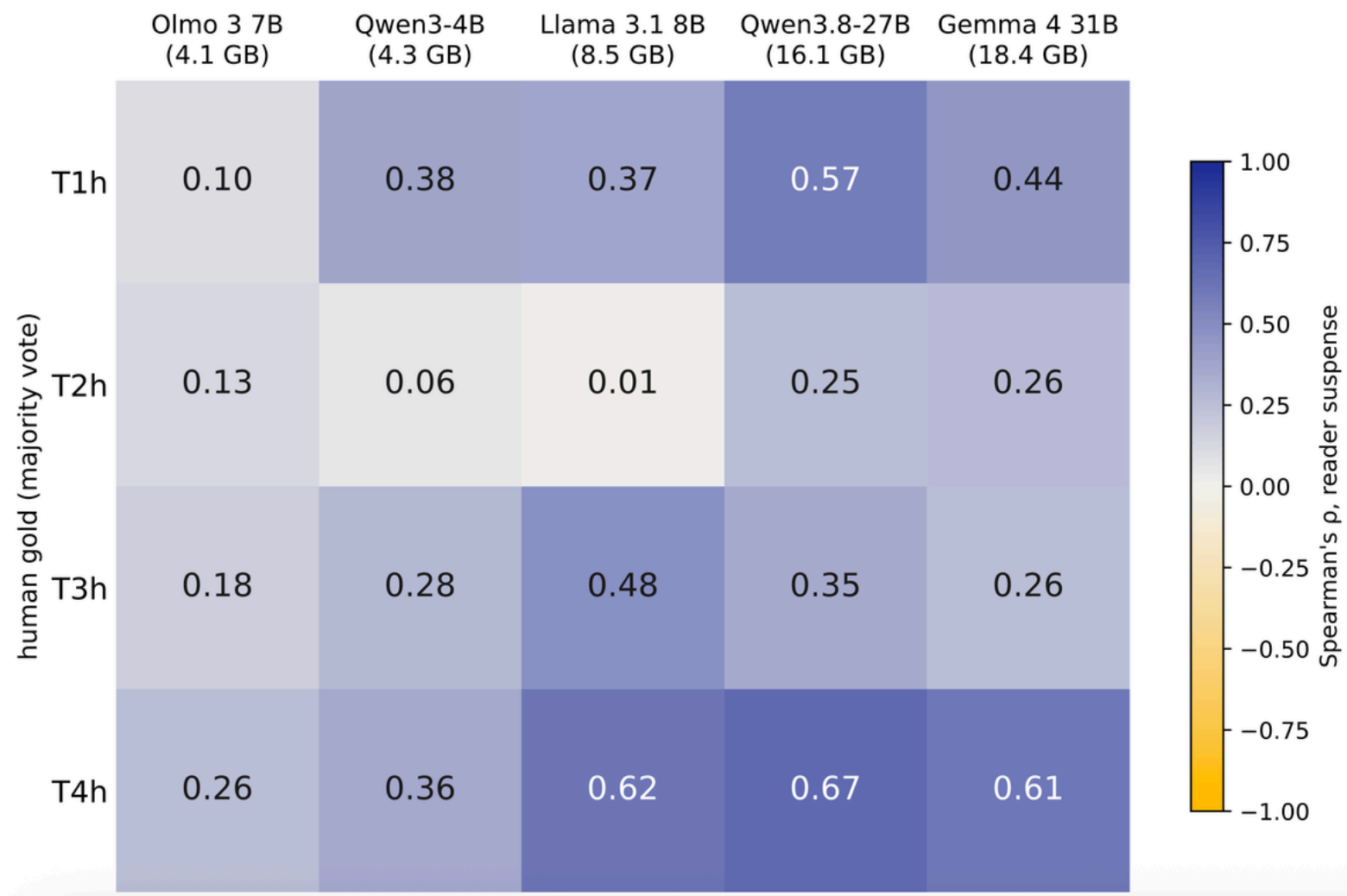


Fig. 3: Human majority vote per theory (rows) against each model prompted with the same theory (columns); Spearman's ρ , reader suspense, models ordered by size.